

Quiz 1

This is an individual assignment.

Question 1 (4 points)

Consider a dataset with the following six samples, where each input sample $x_i \in \{0,1\}$. The target output for each sample x_i is a vector $t_i = [t_{i1}, t_{i2}]^T$, as illustrated in the table below.

Sample x_i	t_{i1}	t_{i2}
$x_1 = 0$	-1	1
$x_2 = 0$	-1	-2
$x_3 = 0$	-1	-1
$x_4 = 1$	1	1
$x_5 = 1$	1	2
$x_6 = 1$	2	1

For example, sample $x_1 = 0$ and its output target value is $t_1 = [t_{11}, t_{12}]^T = [-1, 1]^T$.

Consider the following feature space representation:

$$\phi(x) = \begin{cases} [1, 0]^T & \text{if } x=0 \\ [0, 1]^T & \text{if } x=1 \end{cases}$$

Answer the following questions:

- (0.5 points) Define the feature matrix for this dataset and feature space representation.

Based on the data provided in the table, the feature matrix $\mathbf{X}$ and the target matrix are defined as $\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{t} = \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}$

- (0.5 points) Consider a *Linear Regression* mapper function. Write the general mapper function expression for this feature representation.

Assuming a linear regression problem, the mapper function for this problem is as follows:

$$\begin{aligned} \mathbf{y}(x) &= \mathbf{w}^T \phi(x) = \begin{cases} \mathbf{W}^T \left[\begin{array}{c} 1 \\ 0 \end{array} \right] & \text{if } x=0 \\ \mathbf{w}^T \left[\begin{array}{c} 0 \\ 1 \end{array} \right] & \text{if } x=1 \end{cases} \\ \end{aligned}$$

Note that, since our target variable $\mathbf{t}$ is a column vector 2×1 , then so is our mapper output $\mathbf{y}$. Therefore, the $\mathbf{W}$ must be a matrix vector of shape 2×2 .

3. (1 point) **Consider the Mean Squared Error (MSE) as the objective function and the Learning Algorithm as the Normal Equations. Compute the solution for the parameters $\mathbf{w}$. Show your work.**

We know that the analytical solution for $\mathbf{W}$ through the normal equations is given as:

$$\begin{aligned} \mathbf{W} &= \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{t} \\ \text{Putting it all together we find: } \mathbf{W} &= \left(\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \left(\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 & -2/3 \\ 4/3 & 4/3 \end{bmatrix} \end{aligned}$$

4. (1 point) **How does the solution for $\mathbf{w}$ change if you include the Ridge Regularizer in the objective function with the hyperparameter $\lambda = \frac{1}{3}$?**

The analytical solution would now become:

$$\begin{aligned} \mathbf{W} &= \left(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{t} = \left(\mathbf{X}^T \mathbf{X} + \frac{1}{3} \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{t} \\ \text{Putting it all together we find: } \mathbf{W} &= \left(\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} -1 & 1 \\ -1 & -2 \\ -1 & -1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} & \begin{bmatrix} -1 & 1 & -1 & -2 & -1 & -1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 3 + \frac{1}{3} & 0 & 0 & 3 + \frac{1}{3} & 3 \end{bmatrix}^T \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \\ & \begin{bmatrix} -1 & 1 & -1 & -2 & -1 & -1 & 1 & 1 & 1 & 2 & 2 \\ 1 & \frac{3}{10} & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ & & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -1 & -2 & -1 & -1 & 1 & 1 & 1 & 2 & 2 \\ 1 & \frac{9}{10} & -\frac{6}{10} & \frac{12}{10} & \frac{12}{10} \end{bmatrix} \end{aligned}$$

5. (1 point) Use your solution for $\mathbf{w}$ from (4) to make a prediction for the test samples: $x_7=0$ and $x_8=1$.

$$\begin{aligned} \mathbf{y}(x_7) &= \mathbf{W}^T \phi(x_7) = \begin{bmatrix} -\frac{9}{10} & -\frac{6}{10} & \frac{12}{10} & \frac{12}{10} \end{bmatrix} \\ & \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \mathbf{y}(x_8) &= \mathbf{W}^T \phi(x_8) = \begin{bmatrix} -\frac{9}{10} & -\frac{6}{10} & \frac{12}{10} & \frac{12}{10} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

See below for calculations.

```
In [2]: import numpy as np

W = np.array([[ -9/10, -6/10], [12/10, 12/10]])
W
```

```
Out[2]: array([[ -0.9, -0.6],
               [ 1.2,  1.2]])
```

```
In [4]: y_7 = W.T@np.array([1, 0])
y_8 = W.T@np.array([0, 1])
```

```
In [5]: y_7, y_8
```

```
Out[5]: (array([[ -0.9],
                [-0.6]]),
         array([[ 1.2],
                [ 1.2]]))
```

Question 2 (5 points)

Consider a training dataset $\{(x_i, t_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}$ and $t_i \in \mathbb{R}$.

Answer the following questions:

1. (2 points) **Suppose you are training a mapper function of the form:**

$$y(x) = \exp\left(\sum_{j=0}^M w_j x^j\right)$$

Derive the analytical solution of $\mathbf{w}$ using the Normal Equations and the Mean Squared Error (MSE). Show your work.

Note that we can find an analytical solution only in the log-space of the target variable, i.e.

$$\begin{aligned} y(x) &= \exp\left(\sum_{j=0}^M w_j x^j\right) \quad \ln(y(x)) = \sum_{j=0}^M w_j x^j \\ w_j &= \mathbf{X}^T \mathbf{w} \end{aligned}$$

Now, this is a linear regression problem, and the solution for $\mathbf{w}$ using Normal Equations of the MSE objective function is

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \ln(\mathbf{t})$$

To make predictions, we simply take:

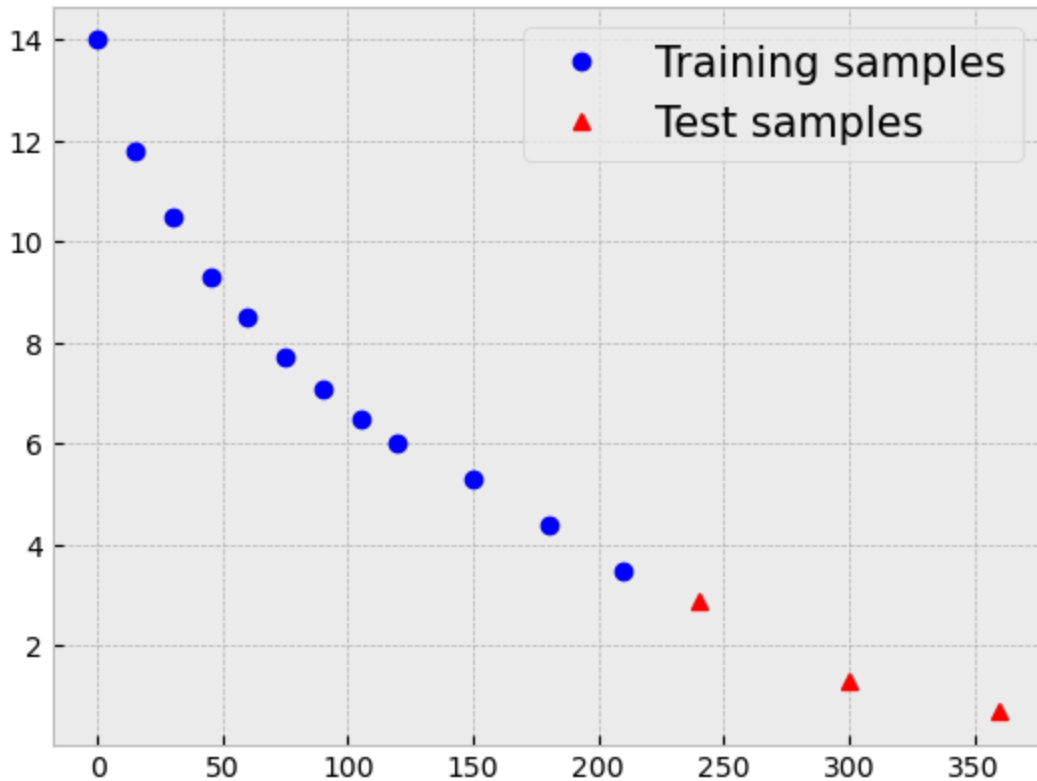
$$y(x) = \exp(\mathbf{X} \mathbf{w})$$

Consider the train sample (blue circles) and test samples (red triangles) below.

```
In [21]: import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
plt.style.use('bmh')

x_train = np.squeeze(np.load('x_train.npy'))
x_test = np.squeeze(np.load('x_test.npy'))
t_train = np.load('t_train.npy')
t_test = np.load('t_test.npy')

plt.plot(x_train, t_train, 'ob', label='Training samples')
plt.plot(x_test, t_test, '^r', label='Test samples')
plt.legend(fontsize=15);
```



3. (2 points) Use the template code from lecture (copied below), modified it as needed to implement your derived solution for mapper function in part (1).

```
In [22]: import scipy.stats as stats
import numpy as np
```

```
In [23]: def exponential_regression(x, t, M):
    # Rename this function and provide the necessary inputs

    # Compute feature matrix
    X = np.array([x**i for i in range(M+1)]).T

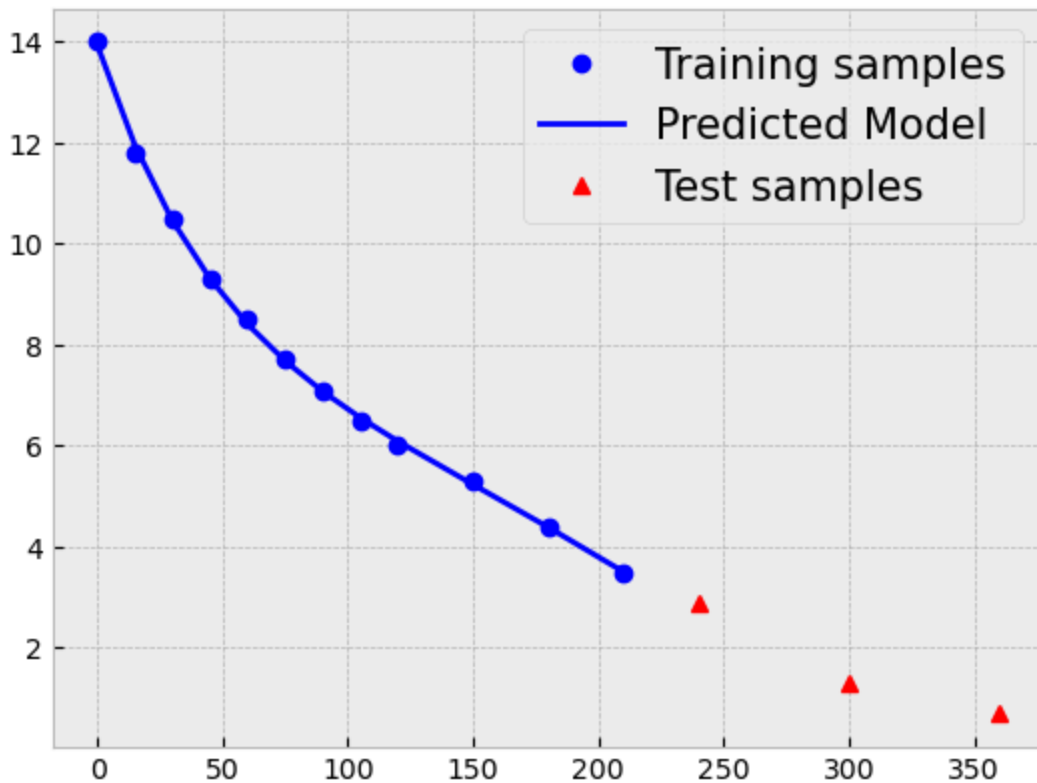
    # Compute the solution for the parameters w
    w = np.linalg.inv(X.T@X)@X.T@np.log(t)

    # Compute model prediction
    y = np.exp(X@w)

    return w, y
```

```
In [26]: #Polynomial order
M = 3 #hard-coding a model order for now
w, y_train = exponential_regression(x_train, t_train, M)

plt.plot(x_train, t_train, 'ob', label='Training samples')
plt.plot(x_train, y_train, 'b', label='Predicted Model')
plt.plot(x_test, t_test, '^r', label='Test samples')
plt.legend(fontsize=15);
```



3. (1 point) **Use your trained models from part (2) to make predictions for the test samples. Discuss your observations.**

```
In [27]: def exponential_regression_test(x, w):
# Rename this function and provide the necessary inputs

# Compute feature matrix
X = np.array([x**i for i in range(len(w))]).T

# Compute model prediction
y = np.exp(X@w)

return y
```

```
In [28]: y_test = exponential_regression_test(x_test, w)
```

```
In [29]: #Polynomial order
M = 3 #hard-coding a model order for now
w, y_train = exponential_regression(x_train, t_train, M)

plt.plot(x_train, t_train, 'ob', label='Training samples')
plt.plot(x_train, y_train, 'b', label='Predicted Model')
plt.plot(x_test, t_test, '^r', label='Test samples')
plt.plot(x_test, y_test, 'r', label='Predicted Model')
plt.legend(fontsize=15);
```

